

COSMETIC PRODUCT ANALYSIS

Preprocessing Steps and Business Questions with Dashboard Visualization

1st PDF Deliverable

Project Name	Cosmetic Product Analysis Dashboard
Tool Used	Tableau / Excel
Dataset	Cosmetic Product Dataset
Team ID	
Date	

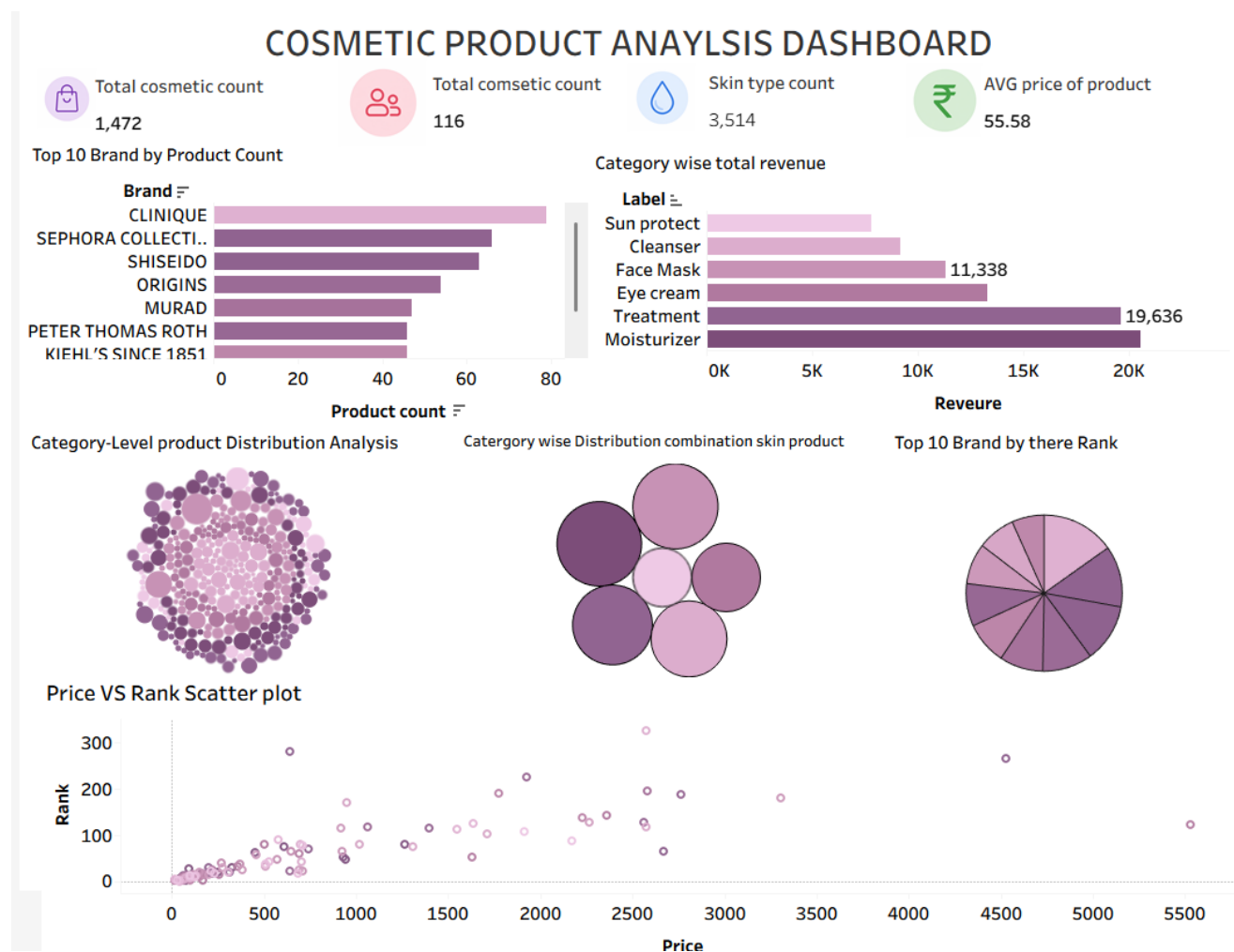


Figure 1: Cosmetic Product Analysis Dashboard uploaded by user

1. Dataset Overview

This project analyzes cosmetic product data to understand product availability, brand performance, category-wise revenue, pricing patterns, skin-type distribution, and product ranking. The final dashboard is created to convert raw cosmetic data into useful business insights for decision-making.

Dashboard KPI	Value / Meaning
Total cosmetic count	1,472 products in the dataset
Total brand count	116 cosmetic brands
Skin type count	3,514 skin-type/product combinations
Average product price	55.58 average price
Main dashboard focus	Product count, revenue, rank, price, category, and brand analysis

2. Preprocessing Steps

Step	Preprocessing Work
1	Imported the cosmetic product dataset into Excel/Tableau.
2	Checked all columns such as brand, label/category, price, rank, skin type, and product name.
3	Removed duplicate records so the same product does not repeat in analysis.
4	Checked missing values in brand, category, price, rank, and skin-type fields.
5	Corrected data types: price and rank were treated as numerical fields; brand and category were treated as dimensions.
6	Standardized inconsistent brand/category names to avoid separate entries for the same brand.
7	Created calculated fields for product count, category revenue, average price, and category distribution.
8	Used filters and sorting to show Top 10 brands, category-wise revenue, and Price vs Rank relationship.
9	Built the final Tableau dashboard using KPI cards, bar charts, bubble charts, pie chart, and scatter plot.

3. Business Questions

Business Question 1: What are the overall KPI values of the cosmetic dataset?

Visualization Used	KPI Cards
Dashboard Image	Updated using uploaded dashboard screenshot
Main Insight	The dashboard shows 1,472 cosmetic products, 116 brands, 3,514 skin-type records/combinations, and an average product price of 55.58. These KPIs give a quick summary of the dataset size and market coverage.



Figure 3: KPI Cards

Business Question 2: Which cosmetic brands have the highest product count?

Visualization Used	Horizontal Bar Chart - Top 10 Brand by Product Count
Dashboard Image	Updated using uploaded dashboard screenshot
Main Insight	CLINIQUE has the highest product count, followed by SEPHORA COLLECTION, SHISEIDO, ORIGINS, MURAD, PETER THOMAS ROTH, and KIEHL'S SINCE 1851. This means these brands have stronger product availability in the dataset.

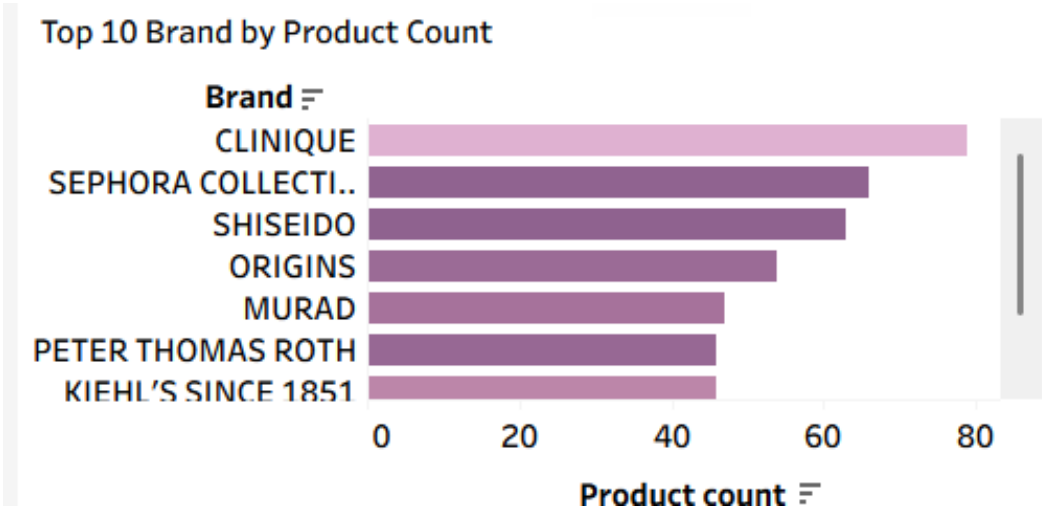


Figure 4: Horizontal Bar Chart - Top 10 Brand by Product Count

Business Question 3: Which product category generates the highest total revenue?

Visualization Used	Horizontal Bar Chart - Category wise total revenue
Dashboard Image	Updated using uploaded dashboard screenshot
Main Insight	Moisturizer and Treatment are the strongest revenue categories. Treatment shows around 19,636 revenue and Eye cream shows around 11,338 revenue. These categories should be prioritized for business planning and promotions.

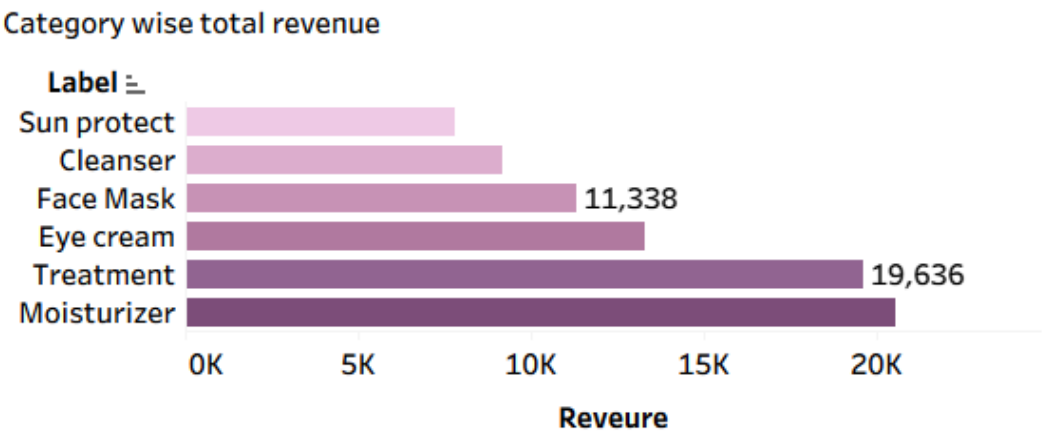


Figure 5: Horizontal Bar Chart - Category wise total revenue

Business Question 4: How are products distributed at the category level?

Visualization Used	Packed Bubble Chart - Category-Level Product Distribution Analysis
Dashboard Image	Updated using uploaded dashboard screenshot
Main Insight	The packed bubble chart shows that cosmetic products are distributed across many categories. Larger bubbles represent categories with more products, while smaller bubbles represent categories with fewer products.

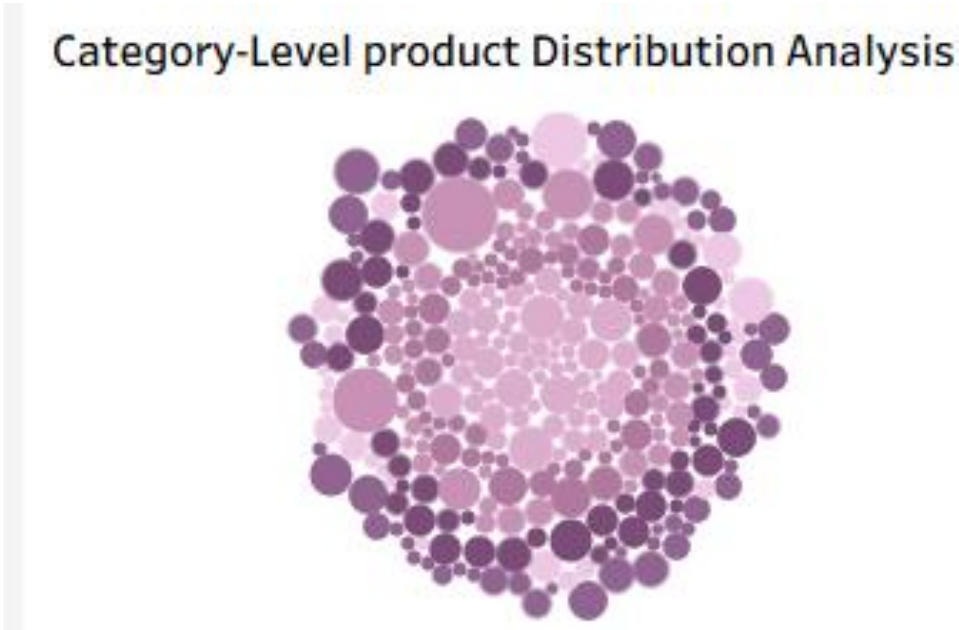


Figure 6: Packed Bubble Chart - Category-Level Product Distribution Analysis

Business Question 5: Which category and skin-product combinations dominate the dataset?

Visualization Used	Bubble Chart - Category wise Distribution combination skin product
Dashboard Image	Updated using uploaded dashboard screenshot
Main Insight	The bubble chart highlights combinations of category and skin products. Larger circles indicate stronger presence in the dataset, helping identify which skin-product groups are more common.

Catergory wise Distribution combination skin product

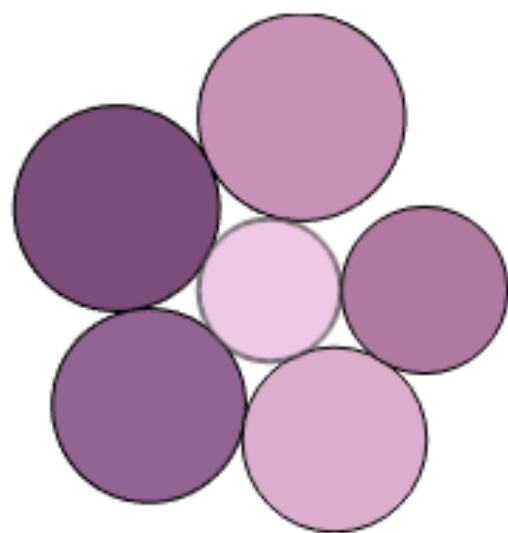


Figure 7: Bubble Chart - Category wise Distribution combination skin product

Business Question 6: How are the top 10 brands distributed by rank?

Visualization Used	Pie Chart - Top 10 Brand by Rank
Dashboard Image	Updated using uploaded dashboard screenshot
Main Insight	The pie chart compares rank share among the top 10 brands. Brands with larger slices have a greater contribution to the rank-based comparison and can be studied for stronger product performance.

Top 10 Brand by there Rank

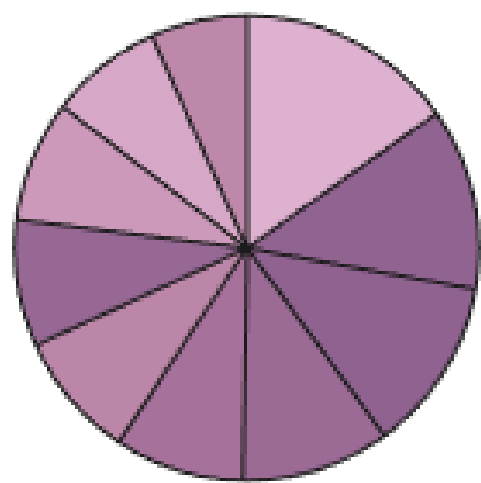


Figure 8: Pie Chart - Top 10 Brand by Rank

Business Question 7: What is the relationship between product price and product rank?

Visualization Used	Scatter Plot - Price VS Rank
Dashboard Image	Updated using uploaded dashboard screenshot
Main Insight	Most products are concentrated in the lower to mid-price range, while a few products have very high prices. The scatter plot helps identify outliers and shows that higher price does not always guarantee a higher rank.

Price VS Rank Scatter plot

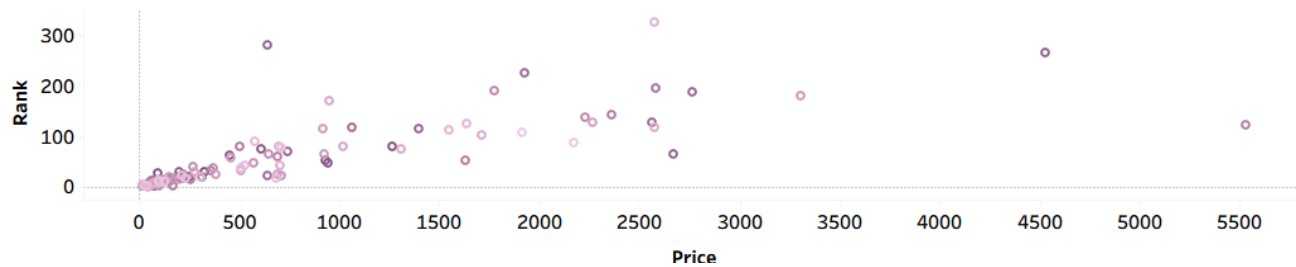


Figure 9: Scatter Plot - Price VS Rank

Business Question 8: Which dashboard insights can support business decisions?

Visualization Used	Combined Dashboard Analysis
Dashboard Image	Updated using uploaded dashboard screenshot
Main Insight	The dashboard suggests that the business should focus on high-revenue categories like Moisturizer and Treatment, track top brands such as CLINIQUE and SEPHORA COLLECTION, and review price-rank outliers to improve product positioning.

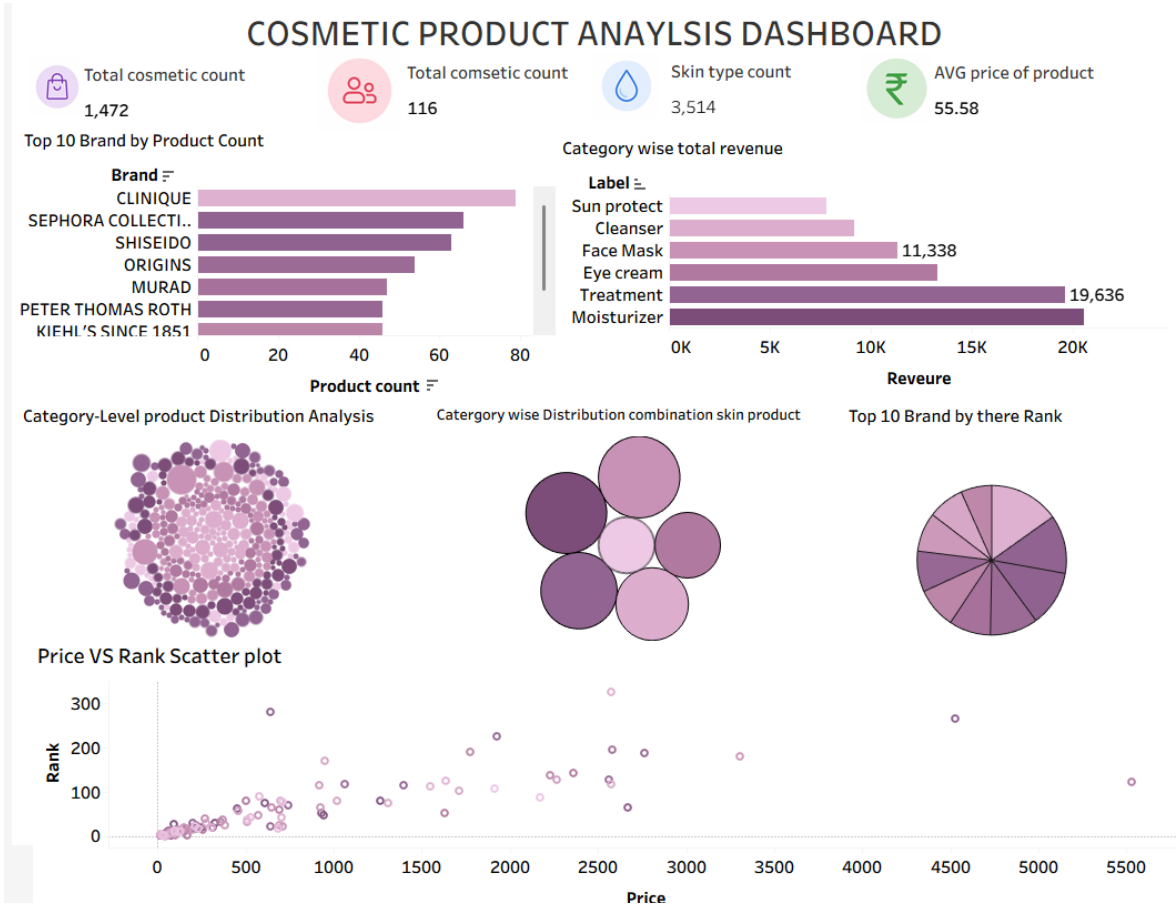


Figure 10: Combined Dashboard Analysis

4. Conclusion

This updated report uses the uploaded Cosmetic Product Analysis Dashboard as the main visualization source. The business questions are now aligned with the actual dashboard charts: KPI cards, top brand product count, category-wise revenue, category distribution, brand rank, and price-rank analysis. The dashboard helps summarize product availability, understand which categories generate revenue, compare brands, and identify pricing/ranking patterns.

5. Future Scope

- Add filters for brand, category, skin type, and price range.
- Add customer reviews and sentiment analysis if review text is available.
- Compare product performance across different price segments.
- Add a recommendation section for best products by skin type and budget.